PersonaScienceBench: Evaluating Scientific Agents with Diverse Simulated Researchers

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Abstract

Scientific-agent benchmarks increasingly test executable research work, but usually provide complete specifications or pair agents with one fixed, cooperative user. In practice, researchers hold undocumented context that an agent must obtain through conversation. We introduce PersonaScienceBench, which holds the scientific task fixed while varying a simulated researcher. We convert 200 tasks from DiscoveryBench, BixBench, EXP-Bench, SciConvBench, and ASI-Bench into collaboration tasks that retain each source's deliverable, grader, and private references. The researcher simulator receives a complete persona profile that is hidden from the actor; a paired persona-off condition uses the same protocol without the identity block. Independent repeated episodes separate collaborator effects from ordinary run variation. Native benchmark scores measure scientific correctness, while interaction metrics measure review, clarification, revision, communication, and language adaptation. Because the persona panel was selected using exploratory pilot outcomes, it characterizes a range of collaborator effects rather than a population average. PersonaScienceBench provides a controlled test of whether scientific-agent performance depends on whom the agent collaborates with.

1 Introduction

Scientific work is collaborative. Language-model agents now perform substantial parts of the scientific workflow. They search literature, propose hypotheses, write and execute analysis code, and assemble results into reports (Lu et al., 2024; Yamada et al., 2025; Gottweis et al., 2025; Mitchener et al., 2025b). The benchmarks used to measure this progress have followed the same trajectory, moving from recall and reasoning questions toward executable analysis tasks drawn from real studies (Majumder et al., 2024; Mitchener et al., 2025a; Kon et al., 2025; Ye et al., 2025). Yet in practice these systems function as co-scientists rather than autonomous investigators, and the case has been made that they are not architecturally built for unsupervised discovery (Bisht et al., 2026). A deployed scientific agent sits inside a collaboration: someone designed the experiment, ran the instrument, and holds the context that determines whether an analysis is valid. That context is frequently undocumented. Which plates were processed on a different day, why three samples were dropped, whether a covariate is confounded with batch, which normalization the lab actually applies—these facts live with the researcher, not in the data directory. The agent's scientific output therefore depends not only on its own reasoning but on an exchange with a particular human who may explain readily, answer only what was literally asked, or misremember a detail.

The evaluation gap. Most scientific agent benchmarks specify each task completely and uniformly, which measures capability under ideal information but says nothing about capability under realistic information transfer. A recent line of interactive benchmarks closes part of this gap by requiring agents to recognize underspecification and ask: HiL-Bench measures whether coding agents escalate at the right moments (Trinh et al., 2026), SciConvBench evaluates multi-turn clarification for ill-posed computational science requests (Somasekharan et al., 2026), and work in tool-agent settings evaluates interaction with a simulated user under domain rules (Yao et al., 2024; Chen et al., 2026; Ta et al., 2026). These benchmarks establish that asking is hard. They do not, however, vary *whom* the agent is asking. The simulated collaborator is typically a single cooperative interlocutor with one disclosure policy, so the collaborator is a fixed property of the environment rather than an experimental condition. The closest departures from this pattern vary designed behavioral axes—non-collaborative user simulators degrade tool agents through impatience, digression, and incomplete utterances (Shim et al., 2025), and evolved persona policies induce realistic behavioral variation in otherwise cooperative

simulators (Chopra et al., 2026)—but do so in retail, airline, and customer service domains where there is no scientific ground truth to get wrong. The result is that three distinct sources of difficulty remain entangled in current evaluation: the intrinsic difficulty of the scientific task, the capability of the agent, and the character of the collaborator the agent happens to be working with. Reporting a single score against a single default collaborator silently averages over the third.

Our approach. This paper introduces PersonaScienceBench, a benchmark that holds the scientific task fixed and varies the simulated researcher. We convert 200 tasks from five scientific agent benchmarks—DiscoveryBench, BixBench, EXP-Bench, SciConvBench, and ASI-Bench—into collaboration tasks, preserving each source's native deliverable and grading rules rather than reducing them to a common format. An actor agent must produce that native deliverable while interacting with a simulated researcher through a shared review protocol. The researcher is conditioned on a complete rendered profile drawn from MatrAIx, a population-scale simulated-user infrastructure whose persona records span 1,290 categorical dimensions covering background, psychology, capability, and behavior (Li et al., 2026a). Our renderer emits the populated attributes as an explicit attribute–value listing grouped by category, so a condition is a fully auditable set of stated attributes rather than authored prose; a single profile typically contributes on the order of two hundred such attributes. The profile is injected verbatim into the researcher simulator's system prompt only, without truncation or projection onto the task domain, so the actor observes personality solely through conversation. Every task is also run with the identical protocol and the identity block removed, giving a paired persona-off control that separates collaborator variation from task difficulty. Evaluator references stay private throughout, and each task–condition cell is repeated across independent episodes so that persona effects can be distinguished from ordinary run-to-run variation. Because the same task is scored under multiple collaborators with a shared baseline, differences are attributable to the collaborator rather than to task selection.

Upstream validation of this persona corpus establishes that conditioning reliably induces declared behavior—91.5% of trials in a controlled adherence study—but does so in Survey, Chatbot, Web, and App environments over a task library concentrated in commerce, software, finance, and healthcare (Li et al., 2026a). Scientific research collaboration is not among those environments, and product feedback rather than scientific correctness is the measured outcome. PersonaScienceBench supplies that missing environment and asks the question it enables: whether persona-induced behavioral variation propagates into graded scientific error.

What we measure. Evaluation is deliberately two-sided. Outcome metrics use each source benchmark's native grader, so a persona effect registers as a change in scientific correctness and not merely as a change in conversational tone. Interaction metrics—review rounds, revisions, questions asked, clarification coverage, message length, and language adaptation—are recorded separately, because a collaborator can change an agent's behavior substantially without changing its score, and can change its score without changing observable behavior. Keeping these separate is what allows a failure under a difficult collaborator to be diagnosed as a failure to obtain available information rather than assumed to be one.

Contributions.

- **A persona-conditioned scientific collaboration benchmark.** We convert 200 tasks from five scientific agent benchmarks into interactive collaboration tasks under a source-preserving contract that retains native deliverables, graders, and private references (Section 3).
- **An experimental design that isolates the collaborator.** Full-profile researcher conditioning, a paired persona-off control on the identical protocol, and repeated independent episodes per cell yield paired comparisons in which the researcher is the manipulated variable (Section 4). We document the two-stage profile selection, including its outcome-dependent final stage, rather than presenting the panel as population-representative.
- **An empirical characterization of collaborator sensitivity.** We report outcome and interaction effects with repeat-based uncertainty, their dependence on task format and scientific field, and evidence-linked conversation cases tracing profile wording to researcher behavior, actor decisions, and graded outcomes (Section 5, Section 6). Completed experiments are reported separately from proposed extensions.

Section 2 positions this work against scientific agent benchmarks, interactive and simulated-user evaluation, and persona simulation. Section 3 defines the benchmark, Section 4 the experimental protocol, Section 5 the results, and Section 6

mechanism evidence from conversation traces. Section 7 states what simulated collaborators and an outcome-selected panel cannot establish.

2 Related Work

2.1 Scientific Agent Benchmarks

Benchmarks for scientific agents have moved from knowledge recall toward executable research work, and the five sources converted in PersonaScienceBench span the range of task objectives this shift produced. DiscoveryBench frames data-driven discovery as hypothesis search and verification over provided datasets, with tasks derived from published findings and graded against reference hypotheses (Majumder et al., 2024). BixBench supplies real bioinformatics analysis capsules and requires agents to execute multi-step computational biology workflows and answer open-ended questions about the data (Mitchener et al., 2025a). EXP-Bench targets the experimental process itself, decomposing research experiments extracted from publications into design and implementation subtasks (Kon et al., 2025). ASI-Bench progressively withdraws methodological guidance across matched conditions to test autonomous execution (Zhou et al., 2026). SciConvBench, discussed further below, is the only one of the five whose native format is dialogue rather than artifact production (Somasekharan et al., 2026). Related efforts scale task collection automatically (Li et al., 2025) or test faithful replication of published analyses (Ye et al., 2025).

These benchmarks differ substantially in deliverable and grading, which motivates our source-preserving conversion contract: reducing them to a common answer format would discard exactly the native scoring that makes a persona effect interpretable as a scientific error. Their information regimes also differ: most use one fixed specification, ASI-Bench varies the amount of methodological guidance, and SciConvBench makes clarification native. None manipulates collaborator identity and behavior while holding the scientific task and scoring procedure fixed. PersonaScienceBench retains these tasks and their graders while making that collaborator the experimental variable.

2.2 Interactive Agents and Simulated Users

A growing body of work evaluates agents that must interact rather than execute a complete specification. τ-bench emulates dynamic conversations between an agent and a language-model user under domain-specific rules (Yao et al., 2024), extended by τ^2-bench to a dual-control setting in which both agent and user can act on the environment (Barrès et al., 2025), and AppWorld-UL constructs tasks that require clarification, confirmation, and infeasibility reporting, simulating the user with explicitly designed knowledge boundaries (Chen et al., 2026). In the science and engineering setting, SciConvBench evaluates multi-turn clarification for ill-posed computational science requests across fluid mechanics, solid mechanics, materials science, and PDE formulation, scoring dialogue quality by rubric-based judge and explicitly requiring no solver, code execution, or tool invocation for scoring (Somasekharan et al., 2026). HiL-Bench isolates the judgment of *when* to escalate, showing that coding agents which succeed given complete context degrade sharply when they must themselves decide whether information is missing (Trinh et al., 2026). Clarification has also been formalized as a sequential decision problem over whether, what, and when to ask (Ta et al., 2026), and user simulation has been used to make agent evaluation more efficient by guiding it toward behavioral diversity (Nakash et al., 2026).

Our work departs from this literature in what is held constant. These benchmarks fix the human and vary the task, measuring whether the agent asks well. PersonaScienceBench fixes the task and varies the human, measuring how much of the scientific outcome is attributable to the collaborator. Two recent lines vary the simulated user deliberately. Non-collaborative user simulators inject requests for unavailable services, digressions, impatience, and incomplete utterances, and show state-of-the-art tool agents degrading substantially under each condition (Shim et al., 2025). Persona Policies argue that LLM-based simulators inherit the cooperative and homogeneous behavior of their base models, and search for persona generator programs that induce realistic behavioral variation while preserving task goals, reporting large fitness gains and improved agent robustness on τ^2-bench retail and airline domains (Chopra et al., 2026). Together these establish the effect we investigate: varying the simulated human changes measured agent performance. Both, however, operate in retail, airline, and customer-service domains, where a failure is a dialogue breakdown or an incorrect transaction, and both vary hand-designed or search-discovered behavioral axes rather than integrated researcher profiles. We instead measure consequences on native scientific graders, where a collaborator effect surfaces as an incorrect hypothesis, a flawed analysis, or an unsubmitted deliverable. User simulation for evaluating interactive systems long predates this literature, from early user models for spoken dialogue evaluation (Eckert et al., 1997) through agenda-based

simulators for dialogue policy learning (Schatzmann et al., 2007) to LLM-based behavioral simulation at scale (Wang et al., 2023).

Validity of the simulator itself. A parallel line of work questions whether LLM user simulators behave as their specification requires, and it bears directly on our design rather than only on our citations. Simulators exhibit *assistant bias*, drifting toward helpful assistant behavior rather than sustaining the user role (Jeong et al., 2026)—a concern for a researcher simulator whose job includes gatekeeping and refusal rather than assistance. Bounding what a simulated user knows is also harder than a prompt instruction suggests, and naive persona instructions do not reliably constrain a simulator's effective knowledge (Balog and Bakken, 2026). Separately, distributing a task across multiple turns can reduce model reliability (Laban et al., 2025), making interaction length a potential confound for protocols with several review rounds. We treat these as threats to the manipulation rather than as background, and Section 7 states how each bounds our claims.

2.3 Persona-Conditioned Simulation

Conditioning language models on personal descriptions to elicit distinct behavior has an established basis. Argyle et al. showed that conditioning on socio-demographic backstories reproduces response distributions of human subgroups, terming this algorithmic fidelity (Argyle et al., 2022). At scale, Persona Hub demonstrated that a large collection of diverse persona descriptions can drive correspondingly diverse model outputs (Ge et al., 2024), and subsequent work studies how much generated diversity persona detail actually produces (Kambhatla et al., 2025). For evaluation specifically, Persona Generators optimize for coverage of the space of possible users rather than density matching of the most probable ones (Paglieri et al., 2026), and frameworks have been proposed to measure whether persona simulation is faithful along adherence, consistency, and naturalness (Bao et al., 2026).

Relation to MatrAIx. The researcher profiles used in PersonaScienceBench are drawn from MatrAIx, a population-scale simulated-user evaluation infrastructure (Li et al., 2026a). Its Persona 8B corpus represents human variation through a shared schema of 1,290 categorical dimensions spanning background, psychology, capability, behavior, and lifestyle. Records are either sampled from a dependency graph that preserves correlated attributes and enforces compatibility rules, or extracted from human-authored sources—Wikipedia biographies, Amazon review histories, the Stack Overflow Developer Survey, the General Social Survey, PRISM Alignment profiles, and consented survey responses—mapped into the same schema and de-identified. The upstream release also attaches natural-language descriptions to human-grounded records, but our renderer does not use them: the injected identity block is an explicit attribute–value listing grouped by category, typically on the order of two hundred populated attributes for a single profile. A condition is therefore a fully auditable set of stated attributes rather than authored prose, which is what allows observable behavior to be checked against exact profile text. A quality-filtered public coreset of approximately one million records is released, comprising 599,847 human-grounded and 400,000 synthetic personas.

MatrAIx also supplies the strongest existing evidence that this form of conditioning works: in a 400-trial controlled adherence study spanning ten behavioral attributes, declared behavior was expressed or correctly suppressed in 91.5% of trials (Li et al., 2026a). That study, however, was conducted in the four environments MatrAIx provides—Survey, AI Chatbot, Web, and App—across a task library concentrated in Commerce, Software, Finance, and Healthcare. Scientific research collaboration is not among them, and product feedback rather than scientific correctness is the measured outcome; related demonstrations of persona-based simulation target the same interactive-application settings (Liu et al., 2026b). PersonaScienceBench extends this line of work in exactly that direction. We take the persona population as given and validated at the level of behavioral adherence, then ask a question the upstream infrastructure does not: whether persona-induced behavioral variation propagates into *graded scientific error* when the simulated human is a research collaborator and the task is scored by a native scientific verifier. Our contribution is therefore the scientific environment, protocol, and outcome measurement, not the persona corpus or its construction.

Caveats adopted from this literature. A review of 63 persona studies found that task and target population are frequently underspecified and that only about a third discuss the representativeness of their personae, proposing a transparency checklist in response (Batzner et al., 2025). Analytic flexibility in simulating human data is substantial enough that defensible configuration choices materially change fidelity conclusions (Cummins, 2025). Work in a scientific domain is directly instructive: when synthetic personas evaluated a genomics visualization system, grounded

personas shifted toward the concerns of documented users while ungrounded ones drifted toward issues real experts never raised, and both missed a preference the expert study observed (Nguyen et al., 2026). The MatrAIx release is itself explicit that its human-grounded records are not population-representative, that calibration targets only four marginal distributions rather than joint coverage over all 1,290 dimensions, and that human studies remain necessary before applying conclusions to real populations (Li et al., 2026a). We therefore treat our profiles as controlled stimuli that induce reproducible interaction conditions, not as substitutes for human researchers, and we do not interpret demographic attributes of a profile as properties of any real group. Additionally, because we inject complete profiles rather than manipulating individual schema dimensions, communication style, availability, and domain knowledge vary together within a condition; Section 7 treats this full-profile confounding directly.

A further distinction from this literature clarifies our architecture. Persona-conditioned simulation separates the relatively persistent *persona*—background, identity, values, skills, personality, enduring preferences—from the more changeable *user state* that many behaviors actually depend on, including a current goal, available knowledge, prior interactions, trust, and frustration (Liu et al., 2026a). Scientific collaboration requires both: a researcher's disposition is persistent, but what they know about *this particular study* is not a property of their biography. PersonaScienceBench separates these sources accordingly. The persona supplies persistent disposition and is the manipulated variable; study-specific context is owned by the task and held identical across persona conditions; and private evaluator references are withheld from the researcher entirely. A profile therefore changes how study context is communicated and reviewed, not what ground truth exists, which is what makes a score difference interpretable as a consequence of collaboration rather than of unequal information endowment.

This separation is imperfect in one respect worth stating. The MatrAIx schema is not purely dispositional: rendered profiles include a small *interaction state* group with fields such as the participant's expertise gap relative to the system, work schedule, and work location. At least one such field is potentially consequential, since a profile that declares its holder an expert testing the system frames the interaction differently from one seeking genuine assistance. These fields are part of the injected condition and are not separately manipulated, so the persona conditions carry a limited amount of state alongside disposition.

2.4 Evaluating Collaboration

Evaluations of interactive agents tend to report either outcome scores or interaction quality, and the two can diverge. Outcome-only scoring cannot distinguish an agent that obtained the information it needed from one that was simply given an easier collaborator; interaction-only scoring, as in rubric-graded clarification dialogues, establishes that an agent asked sensible questions without establishing that the resulting science was correct (Somasekharan et al., 2026). Metrics that combine the two, such as HiL-Bench's pairing of task success with the precision and recall of escalation decisions, come closer by asking whether the agent asked about the things that actually mattered (Trinh et al., 2026).

PersonaScienceBench reports native outcome scores and interaction metrics as separate quantities over the same episodes, with a paired persona-off baseline and repeated independent episodes per cell. This supports three claims that a single aggregate cannot: that a collaborator changed observable agent behavior, that a collaborator changed graded scientific correctness, and that these two need not co-occur. We additionally analyze stage-aligned conversation trajectories and present evidence-linked cases tracing profile wording through researcher behavior to actor decisions and graded outcomes.

Which validity we claim. A fitness-for-purpose framework for persona-conditioned simulation argues that simulated users should be judged against the specific claim they were built to support rather than against general human-likeness, and distinguishes four forms of validity: persona fidelity, whether the model preserves and uses the supplied profile; behavioral and interaction validity, whether actions and trajectories resemble the target human process; task and ecological validity, whether evaluation with simulated users exposes the same capabilities and failures real users would reveal; and distributional validity, whether a collection of simulated users represents an intended population (Liu et al., 2026a). Passing one form does not imply the others, and the framework's central recommendation is to state exactly what has been validated (Li et al., 2026b). We adopt that discipline. PersonaScienceBench inherits persona fidelity from upstream adherence validation (Li et al., 2026a) rather than re-establishing it, and its own contribution is evidence about task validity in a new domain: whether collaborator variation exposes scientific failures that a single default collaborator does not. We explicitly do not claim distributional validity. The final profile panel was selected using pilot outcomes, so

it characterizes the range of collaborator effects rather than estimating an average effect over any researcher population, and we make no claim that behavioral and interaction validity holds against real scientist behavior, which would require human comparison data we do not have. Section 7 develops these constraints.

3 PersonaScienceBench

3.1 Task and Interaction Formulation

A *task* is a scientific problem drawn from a source benchmark, retaining that source's native deliverable and grading contract. An *actor* is the agent under evaluation; it holds the tools, executes code where the track requires it, and produces the deliverable. A *simulated researcher* is a language model that plays the collaborating scientist: it reviews the actor's proposed approach, answers questions, requests revisions, and decides when the work may conclude. A *persona condition* assigns that researcher either a complete rendered profile or, in the control condition, no identity block. An *episode* is one complete interaction from the original task statement to a sealed deliverable. An *evaluator* is the source benchmark's native grader, which runs after the episode is sealed and never participates in the conversation.

Three information domains are kept separate by construction. *Public task context* is supplied identically in every condition and is owned by the task, not the persona; it includes the scientific question, the input files, and the required output format. *Researcher-held information* is what the simulated researcher can supply on request, which creates the reason to interact at all. *Private references* are the evaluator's answer keys and grading rubrics, which are withheld from both actor and researcher throughout. Because study-specific context is task-owned and held fixed, a persona changes how that context is communicated, prioritized, and reviewed, but not what ground truth exists or what information is in principle obtainable. This is what licenses reading a score difference as a consequence of collaboration rather than of unequal information endowment across conditions.

3.2 Scientific Tasks and Source Conversion

Conversion is governed by a source-preserving contract: each task keeps its native deliverable, its native grader, and its native notion of success. Tasks are not rewritten into a common answer format, because doing so would discard the source-specific scoring that makes a persona effect legible as a scientific error rather than as a conversational difference.

Four sources use an execution or design track under a shared active-researcher protocol. DiscoveryBench tasks require the actor to produce a hypothesis specified as context, variables, and relationship, given datasets, metadata, and a discovery goal; its real-world split is manually derived from published papers, while a larger synthetic split is model-generated for controlled difficulty (Majumder et al., 2024). We retain its native hypothesis-match grading, and note that the analysis workflow is an optional output that the native metric does not score, so a workflow produced during collaboration is retained as evidence rather than graded. BixBench tasks place the actor in an empty notebook inside a provisioned environment and require multi-step exploration of heterogeneous input files to answer expert-authored questions; the primary regime is open answer, judged binary-correct against a reference answer rather than by numeric tolerance (Mitchener et al., 2025a).

ASI-Bench requires scientific implementation and execution scored by per-task programmatic gates and weighted scorers against a reference solution, and it natively varies *how much methodological guidance the task supplies*, through matched levels ranging from a full method statement down to an objective alone (Zhou et al., 2026). Our conversion exploits that structure directly, and it is the source in which this benchmark's mechanism is most literally instantiated: the actor receives the objective-only prompt, while the withheld methodological guidance from the next-richer level is assigned to the *researcher* as optional advice the actor can obtain only by asking. Guidance facts are extracted only from explicitly named methodological sections and the richest guidance level is never transferred wholesale, so the researcher holds method-level consultation rather than a complete solution. The resulting condition is objective-only execution with methodological consultation available on request, which is by construction not autonomous objective-only execution, and ASI-Bench numbers reported here are therefore not comparable to native autonomous results at that level. Because the frozen panel draws 40 tasks from a benchmark of roughly 60, source coverage is substantially higher here than for the other four sources.

EXP-Bench natively requires an agent to implement *and execute* a research experiment reconstructed from a published paper, with ground truth spanning design, implementation, and conclusion components (Kon et al., 2025). We convert it

into a **design-only** track, and this is a modification of the benchmark rather than a mode it provides: we retain only its design ground truth and its design-correctness criterion, and the actor produces an experimental design and controlled comparison plan without executing the experiment. Execution-dependent components of the native ground truth are therefore not scored in our conversion, and EXP-Bench results reported here are not comparable to native EXP-Bench leaderboard numbers. Our configuration additionally averages three judge passes per deliverable to reduce grading variance; these are repetitions of *our* grading configuration, not part of the native EXP-Bench protocol, and they are not independent episode trials.

SciConvBench is preserved as a dialogue track rather than being converted into an artifact task. It retains its native ambiguous or internally inconsistent request and its intended-specification split: the actor asks clarification questions or returns a completion signal, with up to ten question turns and a final attempt, and native judges assess specification recovery, intent preservation, and conversational grounding. No scientific solver is executed. Converting SciConvBench into a full-specification copying test would have removed exactly the interactive difficulty it was built to measure.

Source pinning and leakage control. Every source is pinned by revision, and the datasets that ship reference material alongside inputs require explicit leakage control. Source repositories and dataset revisions are recorded by commit hash with per-file SHA-256 checksums, because at least one source has been revised after publication in ways that change measured performance: the BixBench dataset was substantially re-reviewed after its paper appeared, with questions revised for answerability and the schema flattened, so scores are comparable only within a pinned revision and results here are not comparable to the numbers reported in the original paper (Mitchener et al., 2025a). BixBench capsule archives also bundle the analyst's executed solution notebook alongside the input data, which would leak answers directly into the actor's workspace. Our conversion mounts only data members: files whose path components match reference-material patterns are excluded, gold notebooks remain in the immutable archive outside both actor and researcher mounts, and the researcher-visible context is generated from input headers, labels, formats, and identifier examples without reference to any gold notebook, hypothesis, result, or ideal answer. Where a native grader does not verify an execution artifact, this is recorded rather than assumed.

3.3 Sampling and Scientific Coverage

The task panel is a frozen random sample of 200 tasks, 40 from each source, drawn with seed 20260929. The selection manifest is content-addressed by SHA-256 `746191312aa95b39a1b041171b14e0a92dd97d7cb68f6d39d719d6bc6fd20b06533` and the identical panel is used for every condition and every trial; tasks are never resampled between trials.

The sampling policy admits numerical scientific workflows and permits diagnostic plots as intermediate artifacts, while excluding image-centric objectives such as image interpretation, microscopy analysis, reconstruction, computer vision, video, and visual generation. Screening was applied to scientific objectives and required outputs, not only to input file types. Eligibility screening and successful input preparation do not guarantee successful execution or grading, and tasks that fail technically remain visible as outcomes rather than being replaced.

Equal per-source quotas define the design. The resulting distribution over scientific super-families is an empirical consequence of that design, not a field quota: 50 life sciences, 63 physical sciences and engineering, 55 mathematics and computer science, and 32 social sciences and humanities. By answer format the panel contains 80 empirical analyses, 40 designs, 40 specification dialogues, and 40 scientific implementations.

Source and field are therefore partly confounded, and this constrains interpretation throughout. All selected social-science and humanities tasks come from DiscoveryBench, and BixBench dominates the life sciences. Field effects and format effects cannot be fully separated in this panel; comparisons within a single source and format are reported where coverage permits. Subfield, source-study, and task-family tags are retained in the manifest, and multiple tasks may derive from a single source study, so tasks are not treated as independent studies for uncertainty estimation.

3.4 Researcher Profiles and Selection

Profiles are complete rendered MatrAIx persona records (Li et al., 2026a), injected verbatim into the researcher simulator's system identity. No truncation, summarization, or projection onto the task domain is applied, and the same profile text is used for every task and every trial. Neither the actor nor the evaluator receives the profile.

Table 1: Task allocation in the frozen panel. Equal per-source quotas define the design; super-family counts are the empirical distribution of the frozen sample rather than field quotas.

Benchmark	Tasks	Native deliverable
DiscoveryBench	40	Hypothesis (context, variables, relationship)
BixBench	40	Open answer from notebook analysis
EXP-Bench	40	Experimental design (design-only conversion)
SciConvBench	40	Specification recovery dialogue
ASI-Bench	40	Execution with researcher-held guidance
Total	200	

Selection proceeded in two stages, and the second is outcome-dependent. In the first stage, an original sample of 300 profiles was rescreened to 180 eligible records, from which a source-balanced, behaviorally diverse cohort of 30 was selected *without* reference to model scores. An exploratory pilot then ran all 30 profiles plus the persona-off control. In the second stage, ten profiles were selected by *largest absolute deviation* from the persona-off baseline in that pilot—not by highest score—giving, in ranked order, 046, 003, 004, 089, 001, 127, 262, 224, 148, and 026. This panel deliberately contains deviations in both directions. Persona 155 is retained as an additional comparison condition outside the top-ten set and is labeled separately in all analyses.

This selection is disclosed rather than presented as representative. Because the panel was chosen using pilot outcomes, it characterizes the *range* of collaborator effects and must not be read as an unbiased estimate of an average effect over any researcher population; the pilot scores that drove selection are not evidence that these profiles are optimal on new tasks. Profiles are not reselected using results from the expanded campaign.

Two cautions apply to reading these profiles. Full profiles contain many attributes and are sometimes internally inconsistent, so observable behavior must be evaluated against the actual injected text rather than against a demographic label or an assumed expertise level. And because complete profiles are injected, communication style, availability, and domain knowledge covary within a condition; this study does not isolate individual traits or demographic effects (Section 7).

3.5 Collaboration Protocol and Isolation

The four execution and design tracks share an active-researcher protocol in which approach review and result review are mandatory. An episode proceeds in five stages: the researcher requests input inspection and a proposed approach; the actor reports a plan with evidence, assumptions, and uncertainties; the researcher approves the approach or requests a focused clarification; the actor performs the analysis or writes the design and submits a result handoff; and the researcher requests an identifiable revision or concludes, after which the controller seals the deliverable for separate native grading. Each episode allows twelve researcher requests and sixty actor calls. Missing submissions remain explicit outcomes and are not silently excluded.

Because plan and result handoffs are mandatory in this interface, handoffs must not be counted as spontaneous engagement. Extra review requests measure *simulator* activity; questions or new proposals raised inside actor handoffs measure *actor* initiative. These are reported as separate quantities. Shared handoffs contain actor-selected evidence only: the researcher does not automatically inspect private tool logs, so what the researcher knows about the analysis depends on what the actor chose to surface.

One protocol asymmetry is recorded because it affects interpretation. The controller permits termination only during result review, so an unreadable or malformed input can trap a cautious researcher in approach review. This behavior is preserved rather than patched mid-campaign, since altering it would require a new run version and break comparability with completed episodes.

SciConvBench retains its native workflow, with the persona conditioning the simulated user's answers. One consequence is analytically important: the actor's *first* question necessarily precedes any persona-conditioned reply, so changes in actor activity attributable to the persona must be measured only after the first researcher response.

Isolation is enforced on three boundaries. The persona reaches only the researcher simulator. Private evaluator references,

hidden graders, and credentials remain outside both actor and simulator access. General knowledge is available to the simulator in all conditions, including persona-off, so the control differs from the persona conditions only by the presence of the identity block. Episodes execute in isolated environments, and full shared messages, model requests and responses, tool events, submissions, native grades, and termination reasons are recorded for every episode, including failed ones. Legacy optional-question bridge interfaces remain implemented for compatibility but constitute a different interaction condition and are not used here.

4 Experimental Setup

4.1 Research Questions

1. **Behavioral diversity:** Do profiles induce repeatable interaction patterns beyond ordinary run variation?
2. **Performance and fields:** When do profiles help or hinder, and how do effects vary with task format and expertise relationship?
3. **Actor comparison (proposed extension):** Do different actors respond differently to the same collaborator panel?
4. **Conversation cases:** What evidence links profile wording to researcher behavior, actor decisions, and outcomes?

4.2 Models, Conditions, and Repeated Trials

In the main campaign the actor, the researcher simulator, and the configured model judge all use `gpt-6-astra` through the direct OpenAI Responses API with reasoning effort `low`. Reasoning effort `none` was originally intended but is unsupported by this model, so `low` was selected explicitly and held fixed across roles and conditions. Native non-model evaluators remain in place wherever a source benchmark provides one, so model judging is used only where the source itself requires it. The earlier exploratory 30-profile pilot used a different configuration—`GPT-6 Luna` at reasoning effort `none`—and its results are therefore reported as a selection stage rather than pooled with the main campaign.

A trial is one complete pass over the 200-task panel for a set of conditions. Every task–condition cell receives three independent complete episodes, each beginning from the original task with a fresh conversation in which both actor and simulator generate a new trajectory. All three outputs are scored and their mean and within-condition spread are reported; the best of three is never selected, and three endings are never generated from one shared conversation. Within each trial the persona-off baseline is run on the same tasks and its mean is reused as the paired reference for every persona comparison in that trial.

Campaign scope is reported in phases to keep completed work distinct from planned work. The initial campaign covered three conditions—persona off, 148, and 155—at $200 \times 3 \times 3 = 1{,}800$ episodes. A subsequent expansion adds the nine remaining top-deviation profiles at $9 \times 200 \times 3 = 5{,}400$ episodes, reusing the matching persona-off baselines and the existing 148 episodes rather than regenerating them. The top-ten panel plus baseline therefore comprises 6,600 episodes, and 7,200 including the retained 155 condition. These are planned totals.

4.3 Execution and Resource Budgets

Each trial requests 128 GB RAM, 30 CPUs, and a 12-hour Slurm allocation, with up to 30 concurrent episode processes and scientific BLAS/OpenMP thread counts pinned to one to prevent oversubscription. Protocol limits are twelve researcher requests and sixty actor calls per episode in the execution and design tracks, and up to ten question turns plus a final attempt in the SciConvBench track. Per-role output limits and native grading contracts are retained, including EXP-Bench's three judge repetitions, which are grading repetitions rather than independent episode trials.

Expansion batches are submitted as nine sequential 600-episode jobs chained with `afterany` dependencies to avoid overlapping API load, grouped as [046, 003, 004], [089, 001, 127], and [262, 224, 026] and repeated across the three trials. An `afterany` dependency enforces non-overlap but does not certify successful completion, so errors remain visible for later repair rather than being masked by the scheduler.

Provenance is recorded per trial. Trial 1 processed 600 episodes in 3 h 21 min 22 s, yielding 536 scored episodes, 32 missing submissions, and 32 technical errors; three episodes retained from an earlier interrupted campaign are

documented as such. Trials 2 and 3 were submitted fresh with identical task inputs and settings and reuse no episode results. A previously cancelled seven-condition job is not counted as an additional trial.

4.4 Outcome and Interaction Metrics

Outcome metrics are the native score of each source benchmark, the submission and completion rates, the paired score change relative to the persona-off baseline on matched tasks, and the score conditional on submission. Missing submissions are counted as zero in the all-attempt score and are additionally reported separately, so that a persona that suppresses submission entirely is distinguishable from one that produces weak but complete work.

Interaction metrics are recorded separately from outcomes and are not combined into a composite. Collaboration volume is measured by total exchanges, additional approach-review rounds beyond the mandatory first, result revisions, and exhausted dialogue budgets. Actor initiative is measured by questions and new proposals raised inside actor handoffs, kept distinct from researcher-driven extra reviews. In the SciConvBench track the native runner supplies clarification recall (issues asked about over annotated issues), clarification precision (relevant question turns over all question turns), conversation-grounded resolution rate, final resolution rate, silent resolution rate, memory consistency, and intent capture. The conversation-grounded rate is the source's primary success metric, since it requires that annotated issues were both resolved and explicitly clarified with the researcher; the silent rate isolates cases resolved by unstated assumption. Together these distinguish asking more from asking usefully, and resolving through dialogue from resolving without grounding. Because the published per-domain decomposition of final resolution into grounded and silent components does not hold exactly in the source (Somasekharan et al., 2026), each component is reported as exported and none is reconstructed from the others. Communication is measured by visible reply characters per role, language choice, actor language matching, and API token usage, with visible-message length and token usage treated as distinct quantities and whitespace word counts avoided as a multilingual comparison.

For each persona–task pair the field relationship is recorded as direct, adjacent, outside, or unknown, supported by the actual injected profile, with methodological familiarity such as statistics or coding recorded separately. A domain label alone does not establish expertise, and this separation is enforced when interpreting expertise effects.

4.5 Aggregation and Statistical Analysis

Analyses use matched task sets: a task is included for a comparison only when all compared conditions produced an evaluable result, and technical errors are reported separately from substantive failures such as missing submissions. In Trial 1 this matched evaluable set retained 185 of 200 tasks—31 ASI, 40 Bix, 34 Discovery, 40 EXP, and 40 SciConv, with super-family counts of 46 life, 57 physical and engineering, 54 mathematics and computer science, and 28 social sciences and humanities. These are analysis counts for one trial, not replacement sampling quotas, and later trials may exhibit different failure patterns.

Independent repeats are averaged within each task–condition cell before aggregation, and paired baselines are retained rather than collapsed into marginal means. Super-family summaries are task-weighted. Because multiple tasks may derive from a single source study, uncertainty estimation must be source-study aware rather than treating tasks as independent. The multiple-comparison family is predefined for the expanded panel before analysis, and interim task-bootstrap results from the three-condition phase are labeled as interim and are not presented as final repeated-run inference. The existing three-arm confidence-interval script covers only the initial conditions and must not be applied as though it covered all ten personas.

4.6 Optional Extensions

Two extensions are designed but unexecuted and are reported as proposals, not contributions. An *actor comparison* would run additional actor models on the same task and persona panel with simulator, evaluator, tools, and budgets held fixed, reporting mean and worst-persona performance, completion, and whether actor ranking changes across collaborators. A *static reference* would place these results beside conventional non-interactive evaluation on ten tasks: task-owned context supplied upfront, no simulator, evaluator answers still private, and comparable actor budgets. The static reference is a contextual anchor; the primary comparison remains interactive persona-off versus interactive persona-on.

Table 2: Placeholder for benchmark means and 95% confidence intervals. Dashes indicate unpopulated manuscript cells, not zero scores.

Benchmark	Matched tasks	Persona off	Persona 148	Persona 155
ASI-Bench	–	–	–	–
BixBench	–	–	–	–
DiscoveryBench	–	–	–	–
EXP-Bench	–	–	–	–
SciConvBench	–	–	–	–

5 Results

Results are organized so that behavioral claims and outcome claims remain separable. Throughout, comparisons use matched task sets, independent repeats are averaged within each task–condition cell before aggregation, and the paired persona-off baseline from the same trial is the reference. Dashes in tables indicate unpopulated manuscript cells, not zero scores.

5.1 Behavioral Diversity and Repeat Variation

The first question is whether a profile produces a recognizable collaboration pattern at all, and whether that pattern is stable enough across independent episodes to be treated as an effect rather than as noise. Behavior is therefore reported with within-condition variation across the three repeats alongside the between-condition difference, since a large persona difference that is smaller than repeat spread is not evidence of a persona effect.

Two decompositions are enforced. Mandatory handoffs are separated from voluntary exchanges, because the protocol requires plan and result handoffs and counting them as engagement would manufacture an effect present in every condition. And researcher-initiated extra reviews are separated from actor-initiated questions and proposals, because these measure activity by different parties and can move in opposite directions under the same profile.

5.2 Conversation Trajectories

Stage-aligned, role-separated semantic trajectories summarize how conversations diverge over the course of an episode. Two guards apply to their interpretation. Divergence is compared against the distance between independent persona-off repeats, so that ordinary conversational variability provides the reference rather than zero. And embedding distance is neither performance nor fidelity: a profile that shifts language or register will move trajectories without necessarily changing the science, and language effects are labeled explicitly where profiles differ in communication language.

5.3 Performance by Benchmark

Native scores are reported per source rather than pooled, because the sources differ in deliverable, grading contract, and metric scale. Missing submissions are counted as zero in the all-attempt score and also reported separately, so that a profile which suppresses submission is distinguishable from one that yields complete but weaker work.

5.4 Scientific Fields and Expertise Relationships

Field effects are reported at the super-family level, with finer subfield breakdowns only where sample counts support them. Because source and field are partly confounded by construction—all social-science and humanities tasks come from DiscoveryBench, and BixBench dominates the life sciences—field comparisons are made within source and format wherever coverage permits, and cross-family claims are stated as weaker.

Expertise relationship is analyzed as a paired within-persona contrast: for each profile, the task-matched score change from baseline in fields related to its background is compared against the change in unrelated fields. Domain label and methodological familiarity are kept separate, since a medical and statistical background does not by itself constitute computational-biology expertise, and an agronomy background does not extend across the life sciences. Normalization to a 0–100 range does not make these metrics a common calibrated measure, and they are not treated as one.

Table 3: Placeholder for task-weighted super-family means and 95% confidence intervals. Metrics are normalized to 0–100 but are not a common calibrated measure.

Super-family	Matched tasks	Persona off	Persona 148	Persona 155
Life sciences	–	–	–	–
Mathematics and computer science	–	–	–	–
Physical sciences and engineering	–	–	–	–
Social sciences and humanities	–	–	–	–

5.5 Actor Comparison: Proposed Extension

This comparison is designed but unexecuted. The completed campaign evaluates a single actor, so it cannot establish whether collaborator sensitivity generalizes across agents or whether actor rankings are stable across collaborators. No cross-actor conclusion is drawn from single-actor runs.

5.6 Failures and Robustness of Conclusions

Technical failures are reported rather than absorbed. API failures, missing submissions, evaluator incompatibilities, truncation, and repairs are counted separately, and historical and repaired evaluator versions are distinguished so that scores are never silently pooled across grading implementations. Because matched-task analysis excludes any task with a technical error in any compared condition, exclusions can themselves correlate with difficulty; the effect of these exclusions on each conclusion is reported.

6 Discussion and Conversation Case Studies

6.1 From Profile to Interaction to Outcome

Aggregate score differences do not explain themselves. This section presents task-matched cases in which a specific profile instruction can be traced through a researcher message, an actor decision, and the graded deliverable, with the persona-off episode on the same task as the comparison. Each case is built from four linked elements: the exact injected profile wording, short event-linked excerpts from the researcher and actor messages, the subsequent actor action, and the native result alongside its baseline.

Two evidentiary rules apply. Excerpts retain event identifiers so that any claimed link can be checked against the full recorded trace, and annotation is performed with outcome scores hidden from the annotator so that a known score cannot motivate the narrative. Cases are selected to include counterexamples where a profile preference did not produce the expected behavior, not only confirming instances.

6.2 Helpful, Unproductive, and Score-Neutral Changes

Collaborator effects are not monotone, and three categories are worth separating because they have different implications for agent design.

Helpful interventions occur where researcher guidance supplies a constraint or correction the actor had not derived, and the deliverable improves relative to the paired baseline. *Unproductive interactions* consume budget without changing the science: repeated clarification on already-settled points, or approval cycles that terminate in the same deliverable. A protocol asymmetry makes one failure mode structurally available—termination is permitted only during result review, so a cautious researcher facing an unreadable input can stall in approach review—and where this occurs it is identified as a protocol interaction rather than attributed to the profile alone.

Score-neutral behavioral change is the category most easily lost in aggregate reporting. A profile may substantially alter exchange counts, question patterns, message length, or language while leaving the native score statistically indistinguishable from baseline. This is a substantive finding rather than a null result: it demonstrates that behavioral sensitivity and outcome sensitivity are separate properties, and that an agent can adapt its interaction without that adaptation reaching the science.

6.3 Implications for Scientific-Agent Evaluation

Three implications follow from the design if the effects are borne out, and each is stated with its limit.

First, a single benchmark score against one default collaborator is an average over an unreported condition. If collaborator variation moves native scores appreciably, then worst-collaborator performance becomes a property worth reporting alongside mean performance, in the same way that worst-group accuracy is reported in distribution-shift evaluation.

Second, failure under a difficult collaborator is not the same as failure to adapt. Because interaction metrics are recorded separately from outcomes, it is possible to distinguish an agent that never sought available information from one that sought it and was refused, and only the former is an agent deficiency. This distinction is unavailable in outcome-only evaluation and is the practical argument for two-sided measurement. A caveat applies in the other direction: models can perform worse when an equivalent task is distributed across multiple turns rather than posed in one turn (Laban et al., 2025). A persona that provokes more rounds may therefore depress outcomes partly through interaction length, so exchange counts must be read alongside any outcome claim.

Fourth, deference is a distinct failure mode from under-asking, and it is visible only when the collaborator can be wrong. An agent that accepts a researcher's mistaken claim about their own data fails scientifically while appearing maximally cooperative, paralleling documented agent overreliance on unreliable tools (Yang et al., 2026). The present design does not plant false researcher claims, so it measures how agents respond to varied collaborators but not whether they check claims against data evidence. Adding controlled researcher error is the most direct extension, and it would turn collaborator variation into a test of scientific skepticism rather than only of adaptation.

Third, observed associations here are not mechanisms. The design supports claims that conditioning on a profile changed behavior and outcomes on matched tasks; it does not identify which attribute produced the change, because complete profiles vary many dimensions simultaneously. Mechanism claims require single-dimension manipulation, which the underlying schema supports but this study does not perform (Li et al., 2026a). Similarly, nothing here establishes that the same sensitivity would appear with human collaborators; that is a question of ecological validity requiring human comparison data (Liu et al., 2026a).

7 Limitations and Responsible Use

Simulated collaborators, not human scientists. Every researcher in this benchmark is a language model conditioned on a profile. This establishes at most that agents are sensitive to *simulated* collaborator variation. Persona-conditioned simulation distinguishes several forms of validity that do not imply one another (Liu et al., 2026a): we rely on upstream persona-adherence evidence for persona fidelity (Li et al., 2026a), contribute evidence about task validity in a scientific domain, and claim neither behavioral validity against real scientist behavior nor distributional validity over any researcher population. Establishing that simulated researchers elicit the same agent failures real collaborators would elicit requires human comparison data that this study does not collect. Evidence in a neighboring scientific domain is a direct warning: synthetic personas evaluating a genomics visualization system drifted toward concerns real experts never raised and missed a preference the expert study observed (Nguyen et al., 2026).

A small, outcome-selected persona panel. The final ten profiles were chosen by largest absolute deviation from the persona-off baseline in an exploratory pilot. This is an outcome-dependent selection, and its consequences are specific: the panel characterizes the range of collaborator effects rather than estimating a population average, pilot effect sizes are inflated by selection and are reported as a selection stage rather than as effect estimates, and transfer to the expanded task panel is an empirical question rather than an assumption. Pilot and expanded panels may overlap in tasks, and that overlap is disclosed rather than assumed away. Ten profiles plus one control cannot support a factorial model over persona attributes.

Full-profile confounding. Complete profiles are injected verbatim, so communication style, availability, domain knowledge, methodological familiarity, and language vary together within a single condition. A score difference between two personas therefore cannot be attributed to any one dimension. This is a deliberate trade: full profiles preserve the internal coherence of a collaborator and avoid hand-authored trait prompts, at the cost of dimension-level attribution. The underlying schema exposes these dimensions separately (Li et al., 2026a), so single-dimension manipulation is feasible

in principle and is the natural next experiment; it is not performed here. Relatedly, some profiles contain internally inconsistent attributes, so behavior must be read against the exact injected text rather than a coherent imagined persona.

Persona and user state are not the same thing. A profile supplies persistent disposition, not knowledge of the specific study under analysis, and many collaboration-relevant behaviors depend on a changeable user state including current goals, accumulated context, trust, and frustration (Liu et al., 2026a). Study-specific context is task-owned and held identical across conditions by design, which is what makes score differences interpretable, but it also means this benchmark does not model a researcher whose knowledge of their own experiment is partial or mistaken. The separation is also imperfect: rendered profiles include a small interaction-state group, including a field declaring the holder's expertise gap relative to the system, so a limited amount of state is bundled into each persona condition rather than held fixed. Whether persona conditioning persists undiminished across a long multi-round episode is not measured here, and drift within an episode would weaken the manipulation rather than the finding.

Simulator-side threats to the manipulation. Three findings about LLM user simulators bound what our persona conditions can be assumed to hold fixed. First, simulators drift toward assistant behavior rather than sustaining an assigned user role (Jeong et al., 2026); our researcher is required to review, withhold approval, and request revision, so any such drift would compress the difference between persona conditions and the persona-off control in the direction of helpfulness. Second, bounding a simulated user's knowledge is harder than a prompt instruction implies (Balog and Bakken, 2026). Our configuration grants the simulator general knowledge in all conditions, so it does not claim to implement a knowledge-bounded researcher; what a persona changes is disposition and disclosure behavior, not an enforced epistemic boundary, and results should not be read as measuring the consequences of a researcher genuinely not knowing something. Third, models can perform worse when an equivalent task is distributed across multiple turns rather than posed in one turn (Laban et al., 2025), so a persona that induces more review rounds may depress outcomes partly through conversation length rather than through its stated traits; exchange counts are reported alongside outcomes so that this alternative explanation remains visible, but the present design does not separate the two. A related risk is deference: an agent may accept a researcher's incorrect claim rather than checking it against data evidence, a failure mode analogous to documented overreliance on unreliable tools (Yang et al., 2026). We do not systematically plant false researcher claims, so this benchmark does not measure evidence-versus-claim judgment, and that remains a natural extension.

Metric caveats inherited from sources. Native metrics carry documented irregularities that we propagate rather than silently smooth. In the SciConvBench track the conversation-grounded resolution rate is the source's primary success metric, and final resolution decomposes definitionally into grounded and silent resolution; however the published per-domain tables do not satisfy that identity exactly, so components are reported as given and never reconstructed arithmetically from one another (Somasekharan et al., 2026). DiscoveryBench scores the hypothesis text and does not score the analysis workflow, so a persona effect on that source concerns stated conclusions rather than demonstrated analytical soundness (Majumder et al., 2024). Because our EXP-Bench conversion is design-only, its numbers are not comparable to native EXP-Bench results (Kon et al., 2025), and because the BixBench dataset was revised after publication, our numbers are not comparable to the scores in its paper (Mitchener et al., 2025a). Metrics normalized to a common 0–100 range across sources are not a common calibrated measure and are not pooled as one.

Language and domain heterogeneity. Profiles differ in communication language, and at least one selected profile communicates in Mandarin while others use English. Language therefore covaries with profile identity, and metrics sensitive to text volume are affected; visible-message length is reported in characters rather than whitespace words for this reason, but language remains a confound rather than a controlled factor. Domain background likewise varies across profiles in ways that correlate with the field composition of the task panel.

Benchmark and field confounding. Equal per-source quotas produce an unequal and non-orthogonal field distribution. All selected social-science and humanities tasks originate from DiscoveryBench and BixBench dominates the life sciences, so a field effect and a source effect cannot be fully separated. Task format is similarly entangled with source: design-only, dialogue, and execution tracks each come from a single benchmark. Comparisons are therefore reported within source and format where coverage permits, and cross-field claims are correspondingly weaker. Because multiple tasks may derive from one source study, tasks are not statistically independent, and uncertainty estimates must account for source-study clustering.

Three repeats and grading uncertainty. Three independent episodes per task–condition cell bound how precisely within-condition variation can be estimated, particularly for high-variance tasks and for detecting small effects. Where a source benchmark requires a model judge, scores inherit that judge's biases and prompt sensitivity; EXP-Bench's three judge repetitions reduce grading noise but are not independent trials and must not be counted as such. Native non-model evaluators are used wherever available precisely to limit this exposure. Semantic annotations used in case studies are judgments made with outcome scores hidden from the annotator and a subset checked by a second judge, but they remain interpretive evidence rather than measurement.

Technical missingness and single actor. Episodes fail for infrastructure reasons—API errors, truncation, evaluator incompatibility—and these are reported separately from substantive failures such as missing submissions. Matched-task analysis excludes any task with a technical error in any compared condition, which protects pairing but changes the effective panel and can itself correlate with condition difficulty. Sensitivity of conclusions to these exclusions is reported. The completed campaign evaluates a single actor model, so no claim is made about whether other agents exhibit the same collaborator sensitivity or the same ordering across personas; the actor comparison remains an unexecuted proposal.

Provenance, licensing, and privacy. Researcher profiles derive from the MatrAIx persona corpus (Li et al., 2026a), whose human-grounded records are extracted from public sources and de-identified by removing direct identifiers. Exact injected prompts, profile hashes, and renderer versions are published so that conditions are reproducible. Private evaluator references, hidden graders, and credentials are kept outside actor and simulator access and are not released.

Interpretation of persona attributes. These profiles do not represent real people, and no attribute should be read as a property of any demographic group. The upstream release is explicit that human-grounded records are not population-representative, that calibration targets only four marginal distributions rather than joint coverage over all 1,290 schema dimensions, and that human studies remain necessary before applying conclusions to real populations (Li et al., 2026a); broader reviews of persona practice make the same point about representativeness and transparency (Batzner et al., 2025; Cummins, 2025). Results here describe how an agent behaved with a *simulated* collaborator carrying certain stated attributes. They are not evidence about how such an agent would serve, or fail, any real population of scientists, and should not be used to justify claims about groups of researchers or to substitute for evaluation with human collaborators in consequential deployment decisions.

8 Conclusion

PersonaScienceBench evaluates scientific agents against a varying collaborator rather than a fixed one. By converting 200 tasks from five scientific agent benchmarks under a source-preserving contract, conditioning a researcher simulator on complete persona profiles, and pairing every persona condition with an identical persona-off control, it separates a source of difficulty that conventional evaluation leaves entangled with task difficulty and agent capability. Reporting native scientific grades and interaction metrics as distinct quantities makes it possible to say not only whether a collaborator changed the outcome, but whether the agent's behavior changed at all—and these need not coincide.

The practical implication is a change in what a single benchmark number means. An agent's score against one default collaborator is an average over an unstated collaborator condition, and this work supplies the apparatus to measure that dependence directly: worst-collaborator performance, not only mean performance, is a reportable property of a scientific agent.

Three limits bound these conclusions and are not incidental. The collaborators are simulated, so the results concern sensitivity to simulated collaborator variation and not to real scientists. The persona panel was selected using pilot outcomes, so it spans a range of effects rather than estimating an average over researchers. And complete profiles vary many attributes at once, so no effect is attributable to a single dimension.

Each limit indicates a specific next experiment. Single-dimension manipulation is feasible because the underlying schema separates communication, capability, and background attributes (Li et al., 2026a), and would convert range characterization into dimension-level attribution. Evaluating multiple actor models on the same collaborator panel would establish whether collaborator sensitivity is a property of agents in general or of one system, and whether actor rankings are stable across collaborators. Comparison against human collaborators on a small task subset would test the ecological

validity that simulation alone cannot establish (Liu et al., 2026a). The actor comparison and the non-interactive static reference described in Section 4 are designed but unexecuted, and are proposals rather than results of this work.

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A Task Inventory and Conversion Contracts

The frozen panel is a random sample of 200 tasks, 40 from each of five sources, drawn with seed 20260929. The selection manifest is content-addressed by SHA-256 `746191312aa95b39a1b041171b14e0a92dd97d7cb68f6d39d719d6bc6fd20b06533` and is reused identically for every condition and trial; tasks are not resampled between trials. The empirical super-family distribution of the sample is 50 life sciences, 63 physical sciences and engineering, 55 mathematics and computer science, and 32 social sciences and humanities. By answer format the panel contains 80 empirical analyses, 40 experimental designs, 40 specification dialogues, and 40 scientific implementations.

The sampling policy admits numerical scientific workflows and permits diagnostic plots as intermediate artifacts, while excluding image-centric objectives including image interpretation, microscopy and image analysis, reconstruction, computer vision, video, and visual generation. Screening was applied to scientific objectives and required outputs, not only to input file types.

B Full Profiles and Prompt Templates

Profiles are complete rendered MatrAIx persona records (Li et al., 2026a) injected verbatim into the researcher simulator's system identity, with no truncation, summarization, or projection onto the task domain. Selection had two stages: an original 300-profile sample was rescreened to 180 eligible records from which a source-balanced, behaviorally diverse 30-profile cohort was drawn without reference to model scores; ten profiles were then selected by largest absolute deviation from the persona-off baseline in the exploratory pilot, in ranked order 046, 003, 004, 089, 001, 127, 262, 224, 148, and 026. Persona 155 is retained as an additional condition outside that set. The persona-off control uses the same simulator, protocol, and general-knowledge access with the identity block omitted.

C Implementation and Reproducibility

The main campaign uses `gpt-6-astra` via the direct OpenAI Responses API at reasoning effort `low` for the actor, the researcher simulator, and the configured model judge; reasoning effort `none` is unsupported by this model and `low` was selected explicitly. Native non-model evaluators are retained wherever a source provides one. The exploratory 30-profile pilot used `GPT-6 Luna` at reasoning effort `none` and is not pooled with the main campaign.

Each trial requests 128 GB RAM, 30 CPUs, and a 12-hour Slurm allocation, with up to 30 concurrent episode processes and BLAS/OpenMP thread counts pinned to one. Protocol limits are twelve researcher requests and sixty actor calls per episode in the execution and design tracks, and up to ten question turns plus a final attempt in the SciConvBench track.

Initial three-condition trials were submitted as Slurm jobs 1957983 (trial 1), 1962301 (trial 2), and 1962305 (trial 3); trial 1 processed 600 episodes in 3 h 21 min 22 s with 536 scored, 32 missing submissions, and 32 technical errors, and retained three episodes from an earlier interrupted campaign. A cancelled seven-condition job (1957850) is not an additional trial. The nine-profile expansion was submitted as sequential 600-episode batches, jobs 1964501 through 1964513, chained with `afterany` dependencies after resumed trial 3 (job 1964348), grouped as [046, 003, 004], [089, 001, 127], and [262, 224, 026] across trials. An `afterany` dependency enforces non-overlap but does not certify successful completion.

D Extended Outcomes and Statistical Details

Matched-task analysis includes a task only when every compared condition produced an evaluable result. In trial 1 this retained 185 of 200 tasks—31 ASI, 40 Bix, 34 Discovery, 40 EXP, and 40 SciConv—with super-family counts of 46 life, 57 physical and engineering, 54 mathematics and computer science, and 28 social sciences and humanities. These are per-trial analysis counts, not replacement sampling quotas.

Independent repeats are averaged within each task–condition cell before aggregation; paired baselines are retained; super-family summaries are task-weighted. Because multiple tasks may derive from a single source study, uncertainty estimation is source-study aware rather than treating tasks as independent.

E Behavioral Metrics and Annotation Rubric

Metric definitions separate mandatory from voluntary activity, and researcher activity from actor activity. Plan and result handoffs are mandatory in this interface and are not counted as engagement; extra review requests measure simulator activity, while questions and new proposals inside actor handoffs measure actor initiative. The SciConvBench runner exports clarification recall and precision, final resolution rate, conversationally grounded resolution rate, silent resolution rate, memory consistency, and intent capture, where final resolution decomposes into grounded and silent components. Visible-message length is measured in characters by role rather than whitespace words, to remain valid across communication languages, and is kept distinct from API token usage. The actor's first SciConvBench question precedes any persona-conditioned reply, so persona-attributable activity is measured only after the first researcher response.

F Extended Conversation Evidence

G Extended Positioning and Validity

Interactive benchmarks establish that agent performance depends on information exchange, but most use one cooperative user policy. Work on non-collaborative simulators and evolved persona policies shows that impatience, digression, incomplete disclosure, and other user behaviors can alter measured performance (Shim et al., 2025; Chopra et al., 2026). Those evaluations primarily target transactional domains. PersonaScienceBench instead preserves scientific deliverables and native verifiers, making it possible to distinguish a change in conversational behavior from a change in scientific correctness.

Persona-conditioned simulation also raises representativeness concerns. Prior work finds substantial flexibility in persona construction and evaluation, and grounded and ungrounded synthetic users need not surface the same concerns as real experts (Batzner et al., 2025; Cummins, 2025; Nguyen et al., 2026). We therefore use profiles as controlled interaction conditions, not as substitutes for real researchers or demographic groups. Complete profiles vary several attributes together, so this design estimates effects of whole collaborator conditions rather than effects of individual traits.

Following fitness-for-purpose accounts of simulated-user evaluation, we separate persona fidelity, behavioral validity, task validity, and distributional validity (Liu et al., 2026a; Li et al., 2026b). Upstream adherence tests provide evidence that MatrAIx profiles affect declared behavior (Li et al., 2026a); our benchmark tests whether those controlled differences expose changes in scientific-agent behavior and native task outcomes. It does not establish that traces match real scientist behavior or that the outcome-selected profile panel represents a researcher population.

A further class of threats concerns the simulator rather than the profile, and each bounds a different assumption of our design. LLM user simulators drift toward assistant behavior instead of sustaining an assigned role (Jeong et al., 2026), which for a researcher required to review and withhold approval would compress persona conditions toward the cooperative control. Constraining what a simulated user knows is also harder than a prompt instruction implies (Balog and Bakken, 2026); our configuration grants general knowledge in every condition, so a profile alters disposition and disclosure behavior rather than enforcing an epistemic boundary, and results do not measure the consequences of a researcher genuinely lacking a fact. Models can also perform worse when an equivalent task is distributed across multiple turns rather than posed in one turn (Laban et al., 2025), so a profile that provokes more review rounds may depress outcomes partly through conversation length; exchange counts are therefore reported alongside outcomes, although this design does not fully separate the two mechanisms. Finally, deference is distinct from under-asking: an agent that accepts a researcher's incorrect claim fails scientifically while appearing cooperative, paralleling documented overreliance on unreliable tools (Yang et al., 2026). Because we do not plant false researcher claims, evidence-versus-claim judgment is outside the present scope.

H Optional Static Reference